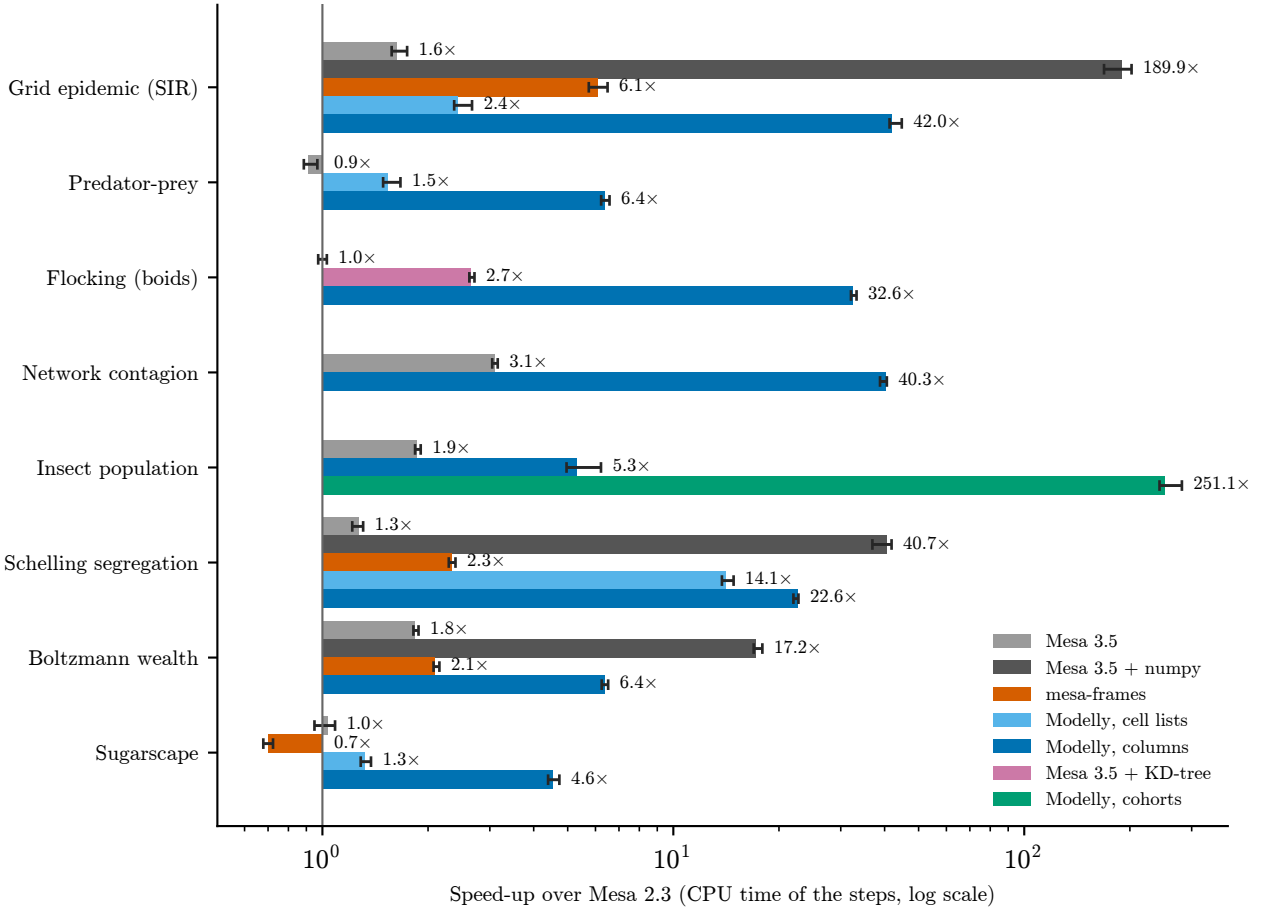
Grid epidemic (SIR)
Predator-prey
Flocking (boids)
Network contagion
Insect population
Schelling segregation
Boltzmann wealth
Sugarscape
1.6×
189.9×
6.1×
2.4×
42.0×
0.9×
1.5×
6.4×
1.0×
2.7×
32.6×
3.1×
40.3×
1.9×
5.3×
251.1×
1.3×
40.7×
2.3×
14.1×
22.6×
1.8×
17.2×
2.1×
6.4×
1.0×
0.7×
1.3×
4.6×
Mesa 3.5
Mesa 3.5 + numpy
mesa-frames
Modelly, cell lists
Modelly, columns
Mesa 3.5 + KD-tree
Modelly, cohorts
10^0
10^1
10^2
Speed-up over Mesa 2.3 (CPU time of the steps, log scale)